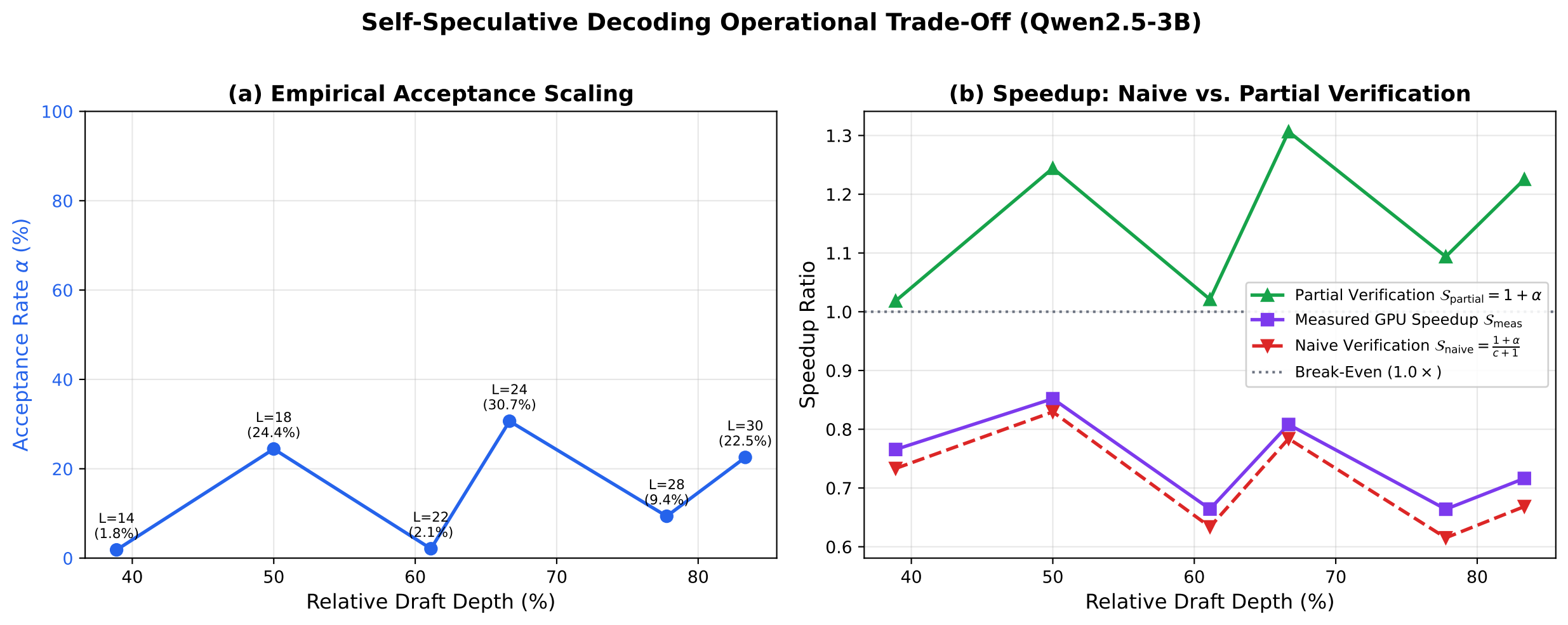
Self-Speculative Decoding Operational Trade-Off (Qwen2.5-3B)
(a) Empirical Acceptance Scaling
Acceptance Rate α (%)
Relative Draft Depth (%)
L=14 (1.8%)
L=18 (24.4%)
L=22 (2.1%)
L=24 (30.7%)
L=28 (9.4%)
L=30 (22.5%)
(b) Speedup: Naive vs. Partial Verification
Speedup Ratio
Relative Draft Depth (%)
Partial Verification $S_{\text{partial}} = 1 + \alpha$
Measured GPU Speedup S_{meas}
Naive Verification $S_{\text{naive}} = \frac{1+\alpha}{c+1}$
Break-Even (1.0 ×)